

The Problem

There is a boundary matrix given

	0	1	2	3	4	5	6	7	8	9	10	11	12
0	0	1	1	1	1	1	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	1	1	0
2	0	0	0	0	0	0	0	1	0	0	1	0	1
3	0	0	0	0	0	0	1	0	0	1	0	0	0
4	0	0	0	0	0	0	0	1	1	1	0	0	0
5	0	0	0	0	0	0	1	0	1	0	0	1	1

and we are making **birth-birth switch forward** transposition (2, 3).

a	b	x	y
2	10	3	8

Computed sets

Successors and Predecessors

	Pred ₁	Pred ₂	Succ ₁	Succ ₂
Set ^{bt} (2, 10)	Pred ₁ ^{bt} (2, 10) = $\emptyset$	Pred ₂ ^{bt} (2, 10) = $\emptyset$	Succ ₁ ^{bt} (2, 10) = $\emptyset$	Succ ₂ ^{bt} (2, 10) = {(3, 8), (4, 7)}
Set ^{bt} (3, 8)	Pred ₁ ^{bt} (3, 8) = {(5, 6), (4, 7)}	Pred ₂ ^{bt} (3, 8) = {(2, 10)}	Succ ₁ ^{bt} (3, 8) = $\emptyset$	Succ ₂ ^{bt} (3, 8) = {(5, 6)}
Set ^{at} (2, 8)	Pred ₁ ^{at} (2, 8) = {(5, 6), (4, 7)}	Pred ₂ ^{at} (2, 8) = {(3, 10)}	Succ ₁ ^{at} (2, 8) = {(3, 10)}	Succ ₂ ^{at} (2, 8) = {(4, 7)}
Set ^{at} (3, 10)	Pred ₁ ^{at} (3, 10) = {(2, 8)}	Pred ₂ ^{at} (3, 10) = $\emptyset$	Succ ₁ ^{at} (3, 10) = $\emptyset$	Succ ₂ ^{at} (3, 10) = {(5, 6), (2, 8)}

$\mathcal{L}$ and $\mathcal{M}$ sets

	Before the Transposition	After the Transposition
$\mathcal{L} = \{(s, t) \in \text{BD} : f(t) < f(y)\}$	{(0, 1), (5, 6), (4, 7)}	{(0, 1), (5, 6), (4, 7)}
$\mathcal{M} = \{(s, t) \in \text{BD} : f(y) < f(t) < f(b)\}$	$\emptyset$	$\emptyset$

Equations

eq	left (formula)	left (para- matrized)	left (value)	right (formula)	right (paramatrized)	right (value)	correct
8	$\text{Succ}_1^{\text{at}}(a, y)$	$\text{Succ}_1^{\text{at}}(2, 8)$	$\{(3, 10)\}$	$\text{Succ}_1^{\text{bt}}(x, y) \oplus \{(x, b)\} \oplus \text{Succ}_1^{\text{bt}}(a, b)$	$\text{Succ}_1^{\text{bt}}(3, 8) \oplus \{(3, 10)\} \oplus \text{Succ}_1^{\text{bt}}(2, 10)$	$\{(3, 10)\}$	✓
9	$\text{Succ}_1^{\text{at}}(x, b)$	$\text{Succ}_1^{\text{at}}(3, 10)$	$\emptyset$	$\text{Succ}_1^{\text{bt}}(a, b)$	$\text{Succ}_1^{\text{bt}}(2, 10)$	$\emptyset$	✓
10	$\text{Pred}_1^{\text{at}}(a, y)$	$\text{Pred}_1^{\text{at}}(2, 8)$	$\{(5, 6), (4, 7)\}$	$\text{Pred}_1^{\text{bt}}(x, y)$	$\text{Pred}_1^{\text{bt}}(3, 8)$	$\{(5, 6), (4, 7)\}$	✓
11	$\text{Pred}_1^{\text{at}}(x, b)$	$\text{Pred}_1^{\text{at}}(3, 10)$	$\{(2, 8)\}$	$\text{Pred}_1^{\text{bt}}(a, b) \oplus \{(a, y)\}$	$\text{Pred}_1^{\text{bt}}(2, 10) \oplus \{(2, 8)\}$	$\{(2, 8)\}$	✓
12	$\text{Succ}_2^{\text{at}}(a, y)$	$\text{Succ}_2^{\text{at}}(2, 8)$	$\{(4, 7)\}$	$\text{Succ}_2^{\text{bt}}(x, y) \oplus \{(x, b), (a, b)\}$	$\text{Succ}_2^{\text{bt}}(3, 8) \oplus \{(3, 10), (2, 10)\}$	$\{(2, 10), (5, 6), (3, 10)\}$	✗
13	$\text{Succ}_2^{\text{at}}(x, b)$	$\text{Succ}_2^{\text{at}}(3, 10)$	$\{(5, 6), (2, 8)\}$	$\text{Succ}_2^{\text{bt}}(a, b)$	$\text{Succ}_2^{\text{bt}}(2, 10)$	$\{(3, 8), (4, 7)\}$	✗
14	$\text{Pred}_2^{\text{at}}(a, y)$	$\text{Pred}_2^{\text{at}}(2, 8)$	$\{(3, 10)\}$	$[\text{Pred}_2^{\text{bt}}(a, b) \cap \mathcal{L}]$	$[\text{Pred}_2^{\text{bt}}(2, 10) \cap \mathcal{L}]$	$\emptyset$	✗
15	$\text{Pred}_2^{\text{at}}(x, b)$	$\text{Pred}_2^{\text{at}}(3, 10)$	$\emptyset$	$\text{Pred}_2^{\text{bt}}(x, y) \oplus \{(a, y)\} \oplus$ $[\text{Pred}_2^{\text{bt}}(a, b) \cap \mathcal{M}]$	$\text{Pred}_2^{\text{bt}}(3, 8) \oplus \{(2, 8)\} \oplus$ $[\text{Pred}_2^{\text{bt}}(2, 10) \cap \mathcal{M}]$	$\{(2, 10), (2, 8)\}$	✗